A. Guess a number!

time limit per test 1 second
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1400 | Tags: greedy, implementation, two pointers | https://codeforces.com/problemset/problem/416/A

A TV show called "Guess a number!" is gathering popularity. The whole Berland, the old and the young, are watching the show.

The rules are simple. The host thinks of an integer *y* and the participants guess it by asking questions to the host. There are four types of acceptable questions:

- Is it true that *y* is strictly larger than number *x*?
- Is it true that *y* is strictly smaller than number *x*?
- Is it true that *y* is larger than or equal to number *x*?
- Is it true that *y* is smaller than or equal to number *x*?

On each question the host answers truthfully, "yes" or "no".

Given the sequence of questions and answers, find any integer value of *y* that meets the criteria of all answers. If there isn't such value, print "`Impossible`".

Input

The first line of the input contains a single integer *n* (1 ≤ *n* ≤ 10000) — the number of questions (and answers). Next *n* lines each contain one question and one answer to it. The format of each line is like that: "*sign x answer*", where the *sign* is:

- ">" (for the first type queries),
- "<" (for the second type queries),
- ">=" (for the third type queries),
- "<=" (for the fourth type queries).

All values of *x* are integer and meet the inequation - 10^9 ≤ *x* ≤ 10^9. The *answer* is an English letter "Y" (for "yes") or "N" (for "no").

Consequtive elements in lines are separated by a single space.

Output

Print any of such integers *y*, that the answers to all the queries are correct. The printed number *y* must meet the inequation - $2 \cdot 10^9$ ≤ *y* ≤ $2 \cdot 10^9$. If there are many answers, print any of them. If such value doesn't exist, print word "`Impossible`" (without the quotes).

Examples

Input
4 >= 1 Y < 3 N <= -3 N > 55 N

Output

```
17
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Input

```
2
> 100 Y
< -100 Y
```

Output

```
Impossible
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